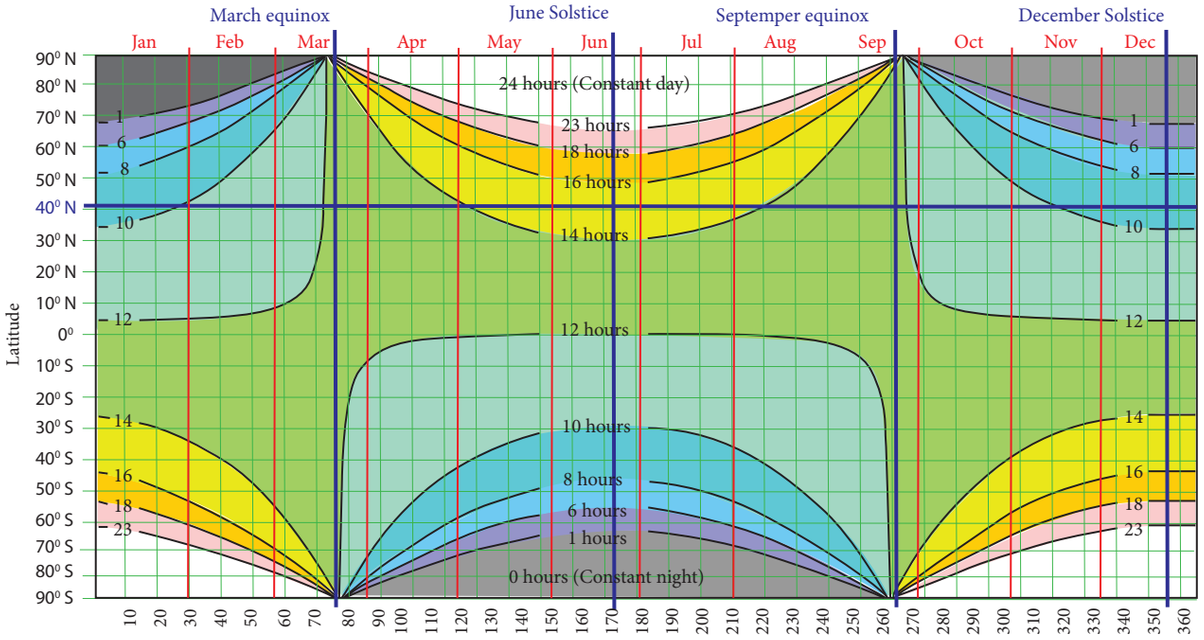


Figure 2.13 Variations in the Length of Day and Night

Table 2.2 Variation in the length of day time

Latitude	Summer Solstice	Winter Solstice	Equinoxes
0°	12 hrs	12 hrs	12hrs
10°	12hrs 35 min	11hrs 25 min	12hrs
20°	13hrs 12min	10hrs 48min	12hrs
30°	13hrs 56min	10hrs 4 min	12hrs
40°	14hrs 52 min	9 hrs 8 min	12hrs
50°	16hrs 18min	7 hrs 42 min	12hrs
60°	18hrs 27min	5 hrs 33min	12hrs
70°	24 hrs (for 2 months)	0 hrs 00 min	12hrs
80°	24 hrs (for 4 months)	0 hrs 00 min	12hrs
90°	24 hrs (for 6 months)	0 hrs 00 min	12hrs

the nights are short here. In the northern hemisphere the nights are longer than the days at this time. The southern hemisphere has summer. The northern hemisphere has winter. This day (22 December), when the sun's rays fall vertically on the Tropic of Capricorn, is known as winter solstice.

Eclipses

Let us understand the effect of the revolution of the earth on the length of the days and the nights. The duration of the daylight varies with latitude and seasons.

An eclipse is a complete or partial obscuration of light from a celestial body and it passes through the shadow of another celestial body. The eclipses are of two types. They are:

A) Solar Eclipse

It occurs on New Moon days, when the moon is between the Sun and the Earth. Thus it obscures a part of the Sun viewed from the Earth, but only from a

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